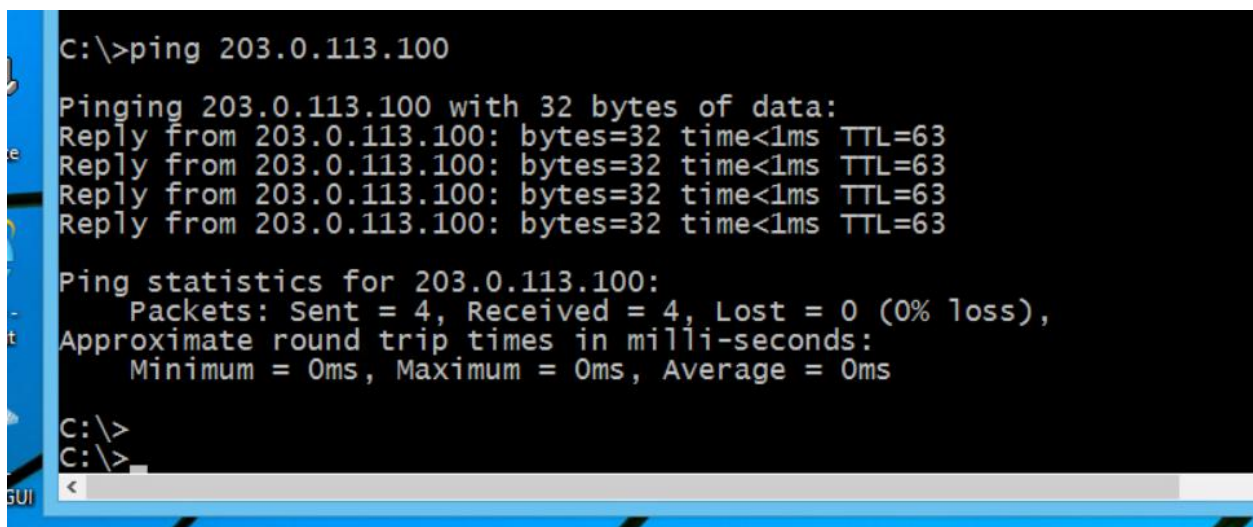


## Lab 2.1: Securing the pfSense Firewall

1. Describe what the command **ipconfig** does

**Ipconfig lists all the network details that the machine is connected to.**

2. Question 3



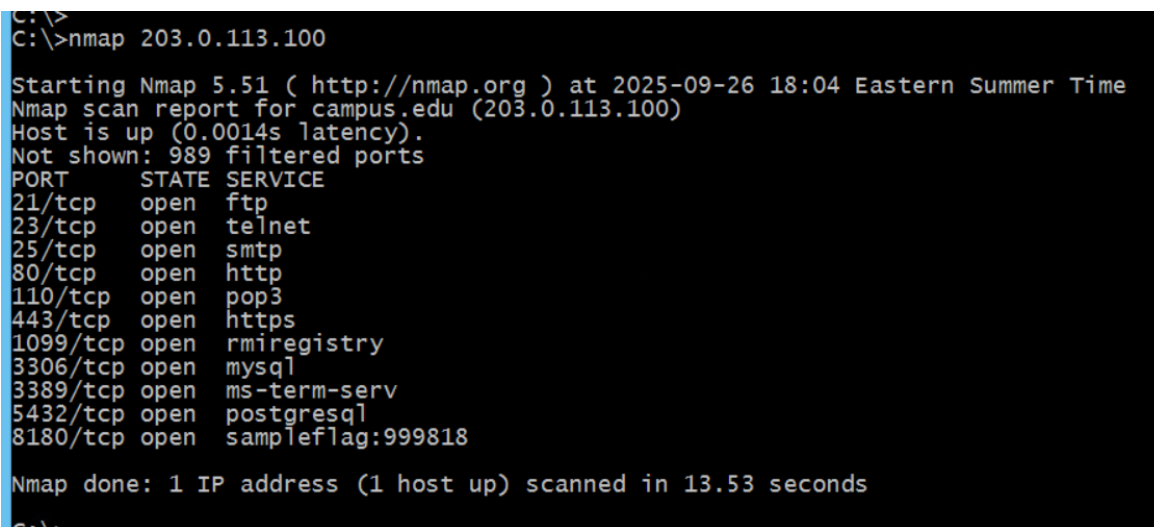
```
C:\>ping 203.0.113.100

Pinging 203.0.113.100 with 32 bytes of data:
Reply from 203.0.113.100: bytes=32 time<1ms TTL=63
Reply from 203.0.113.100: bytes=32 time<1ms TTL=63
Reply from 203.0.113.100: bytes=32 time<1ms TTL=63
Reply from 203.0.113.100: bytes=32 time<1ms TTL=63

Ping statistics for 203.0.113.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
C:\>
```

3. Question 4



```
C:\>
C:\>nmap 203.0.113.100

Starting Nmap 5.51 ( http://nmap.org ) at 2025-09-26 18:04 Eastern Summer Time
Nmap scan report for campus.edu (203.0.113.100)
Host is up (0.0014s latency).
Not shown: 989 filtered ports
PORT      STATE SERVICE
21/tcp    open  ftp
23/tcp    open  telnet
25/tcp    open  smtp
80/tcp    open  http
110/tcp   open  pop3
443/tcp   open  https
1099/tcp  open  rmiregistry
3306/tcp  open  mysql
3389/tcp  open  ms-term-serv
5432/tcp  open  postgresql
8180/tcp  open  sampleflag:999818

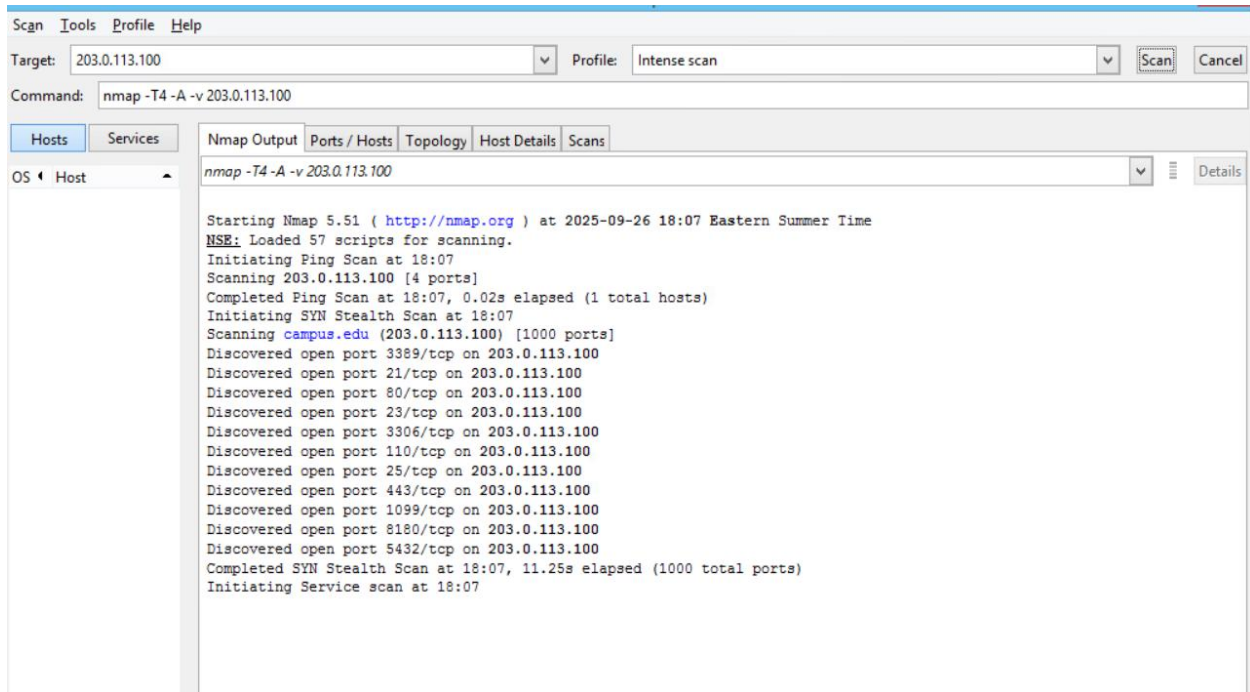
Nmap done: 1 IP address (1 host up) scanned in 13.53 seconds

C:\>
```

4. Research and describe what **Zenmap** does and some advantages compared to **Nmap**

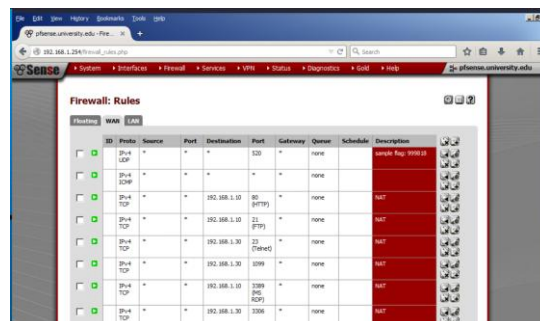
**Zenmap is the GUI for Nmap that simplifies scanning by offering saved profiles, visual network maps, and easier result comparison.**

5. Question 6



## Lab 2.1: Part 2 – Closing Unnecessary Ports on the pfSense Firewall

1. Question 6



## 2. Question 12

Firewall: NAT: Port Forward

The changes have been applied successfully. You can also monitor the filter reload progress.

Port Forward 1:1 Outbound NPT

|                          | If  | Proto | Src. addr | Src. ports | Dest. addr  | Dest. ports   | NAT IP       | NAT Ports     | Description |
|--------------------------|-----|-------|-----------|------------|-------------|---------------|--------------|---------------|-------------|
| <input type="checkbox"/> | WAN | TCP   | *         | *          | WAN address | 443 (HTTPS)   | 192.168.1.10 | *             |             |
| <input type="checkbox"/> | WAN | TCP   | *         | *          | WAN address | 3389 (MS RDP) | 192.168.1.10 | 3389 (MS RDP) |             |

pass  
linked rule

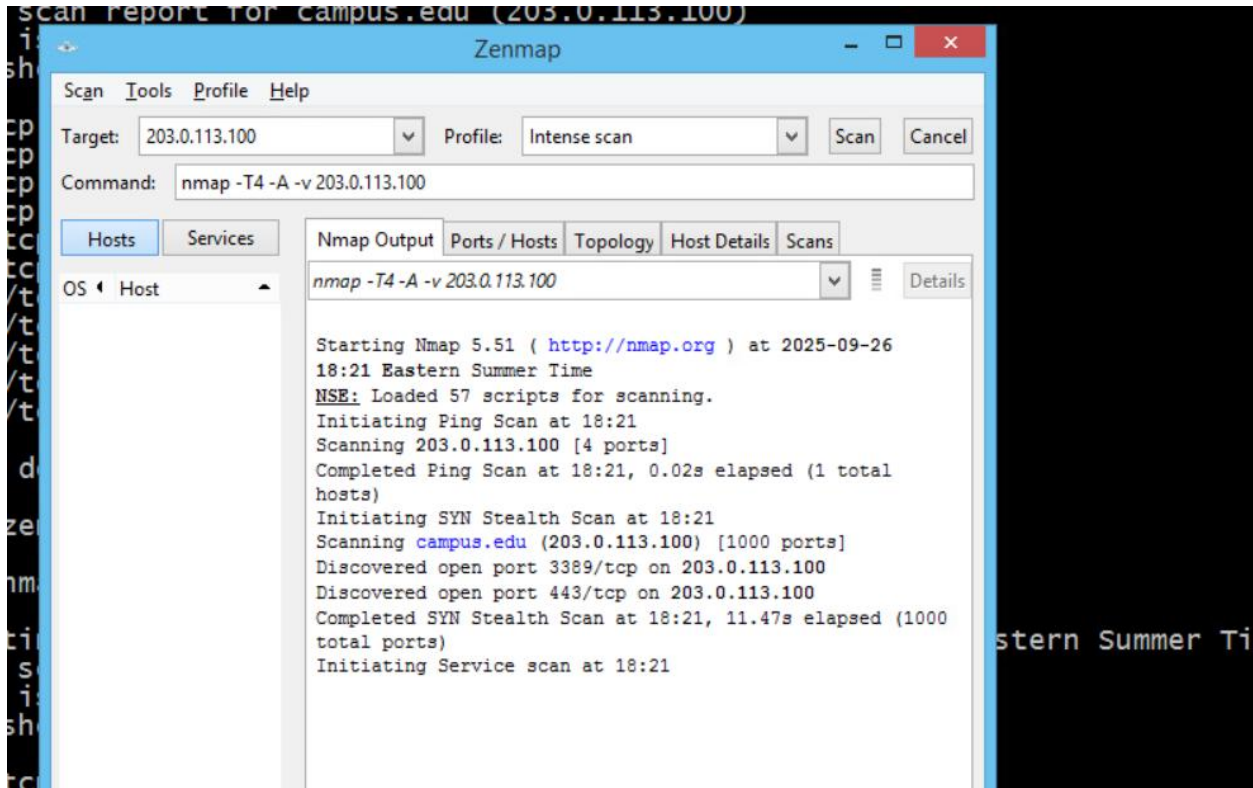
## 3. Question 15

```
C:\>nmap 203.0.113.100

Starting Nmap 5.51 ( http://nmap.org ) at 2025-09-26 18:20 Eastern Summer Time
Nmap scan report for campus.edu (203.0.113.100)
Host is up (0.00s latency).
Not shown: 998 filtered ports
PORT      STATE SERVICE
443/tcp   open  https
3389/tcp   open  ms-term-serv

Nmap done: 1 IP address (1 host up) scanned in 12.84 seconds
C:\>
```

4. Question 17



5. Why did just those two ports show up in the **Zenmap** and **nmap** scan?

**Only those two ports (443 and 3389) showed up in the Zenmap/Nmap scan because they were the only ports that responded as open on the target system.**

## Lab 2.1: Part 3 – Adding a Secure Service to the pfSense Firewall

1. Research and describe what a **forwarding port rule** does

A forwarding port rule is a network configuration that directs traffic arriving on a specific port of one device

2. Question 6

**Firewall: NAT: Port Forward**

**The NAT configuration has been changed. You must apply the changes in order for them to take effect.** [Apply changes](#)

Port Forward 1:1 Outbound NAT

|                          | If  | Proto | Src. addr | Src. ports | Dest. addr  | Dest. ports   | NAT IP       | NAT Ports     | Description |  |
|--------------------------|-----|-------|-----------|------------|-------------|---------------|--------------|---------------|-------------|--|
| <input type="checkbox"/> | WAN | TCP   | *         | *          | WAN address | 22 (SSH)      | 192.168.1.30 | 22 (SSH)      |             |  |
| <input type="checkbox"/> | WAN | TCP   | *         | *          | WAN address | 443 (HTTPS)   | 192.168.1.10 | *             |             |  |
| <input type="checkbox"/> | WAN | TCP   | *         | *          | WAN address | 3389 (MS RDP) | 192.168.1.10 | 3389 (MS RDP) |             |  |

pass  
 linked rule

### 3. Question 7

**Firewall: NAT: Port Forward**

 The changes have been applied successfully. You can also monitor the filter reload progress. Close

Port Forward **1:1** Outbound NPt

|                          | If                                                                                    | Proto | Src. addr | Src. ports | Dest. addr  | Dest. ports   | NAT IP       | NAT Ports     | Description |                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------|---------------------------------------------------------------------------------------|-------|-----------|------------|-------------|---------------|--------------|---------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> |  WAN | TCP   | *         | *          | WAN address | 22 (SSH)      | 192.168.1.30 | 22 (SSH)      |             |     |
| <input type="checkbox"/> |  WAN | TCP   | *         | *          | WAN address | 443 (HTTPS)   | 192.168.1.10 | *             |             |     |
| <input type="checkbox"/> |  WAN | TCP   | *         | *          | WAN address | 3389 (MS RDP) | 192.168.1.10 | 3389 (MS RDP) |             |     |

 pass  
 linked rule

### 4. Question 10

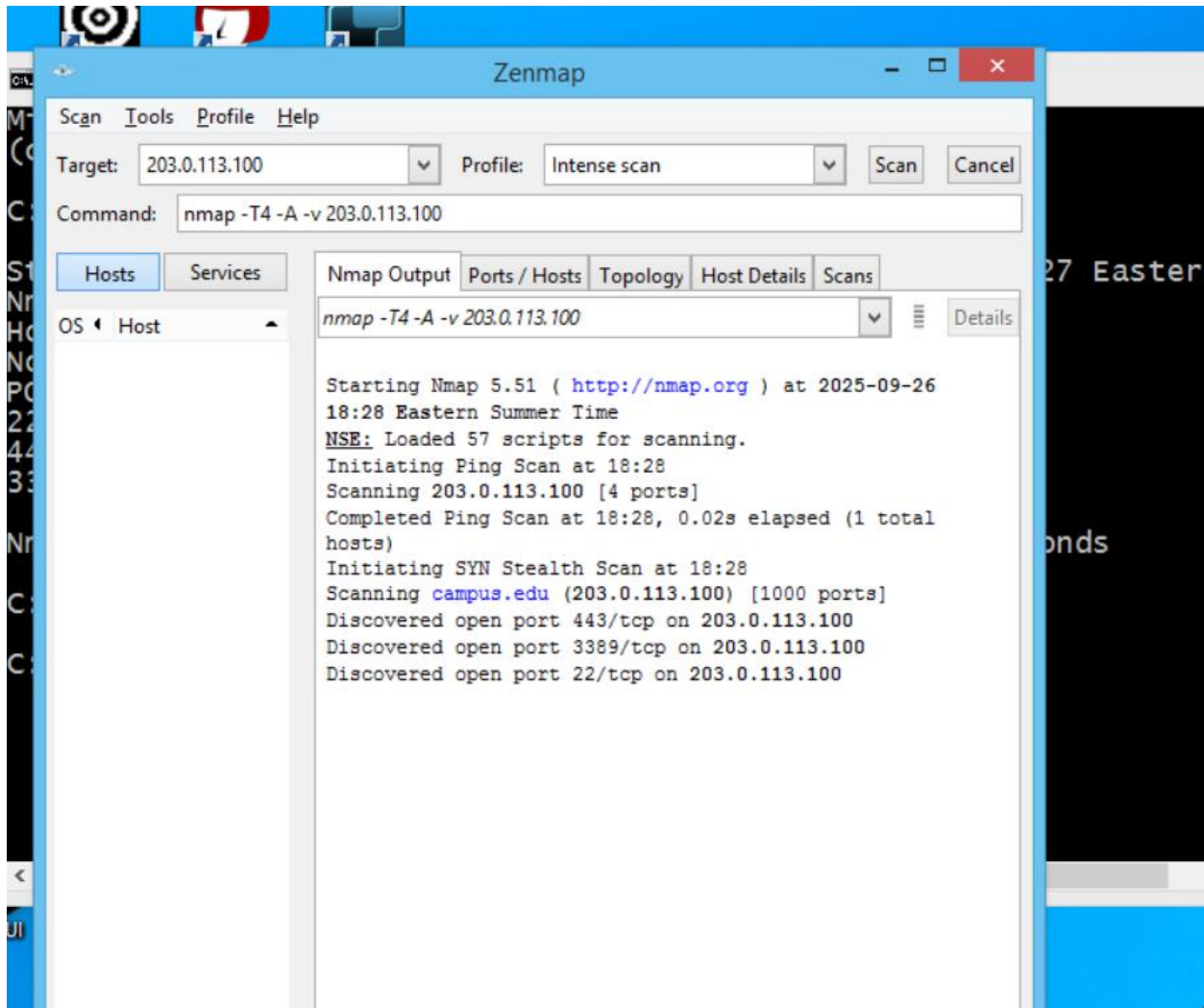
```
cmd - Shortcut
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\>nmap 203.0.113.100

Starting Nmap 5.51 ( http://nmap.org ) at 2025-09-26 18:27 Eastern Summer T
Nmap scan report for campus.edu (203.0.113.100)
Host is up (0.00s latency).
Not shown: 997 filtered ports
PORT      STATE SERVICE
22/tcp    open  ssh
443/tcp    open  https
3389/tcp   open  ms-term-serv

Nmap done: 1 IP address (1 host up) scanned in 12.84 seconds
C:\>
```

5. Question 12



6. What ports showed up in the **Zenmap** and **nmap** scan?

The Zenmap and Nmap scan of the target host 203.0.113.100 revealed three open ports. Port 443/tcp was detected, which is commonly associated with HTTPS traffic and indicates that the system is running a secure web service. Port 3389/tcp was also open, suggesting that Remote Desktop Protocol (RDP) is enabled, allowing remote graphical access to the machine.



7. Research and describe what **PuTTY** is

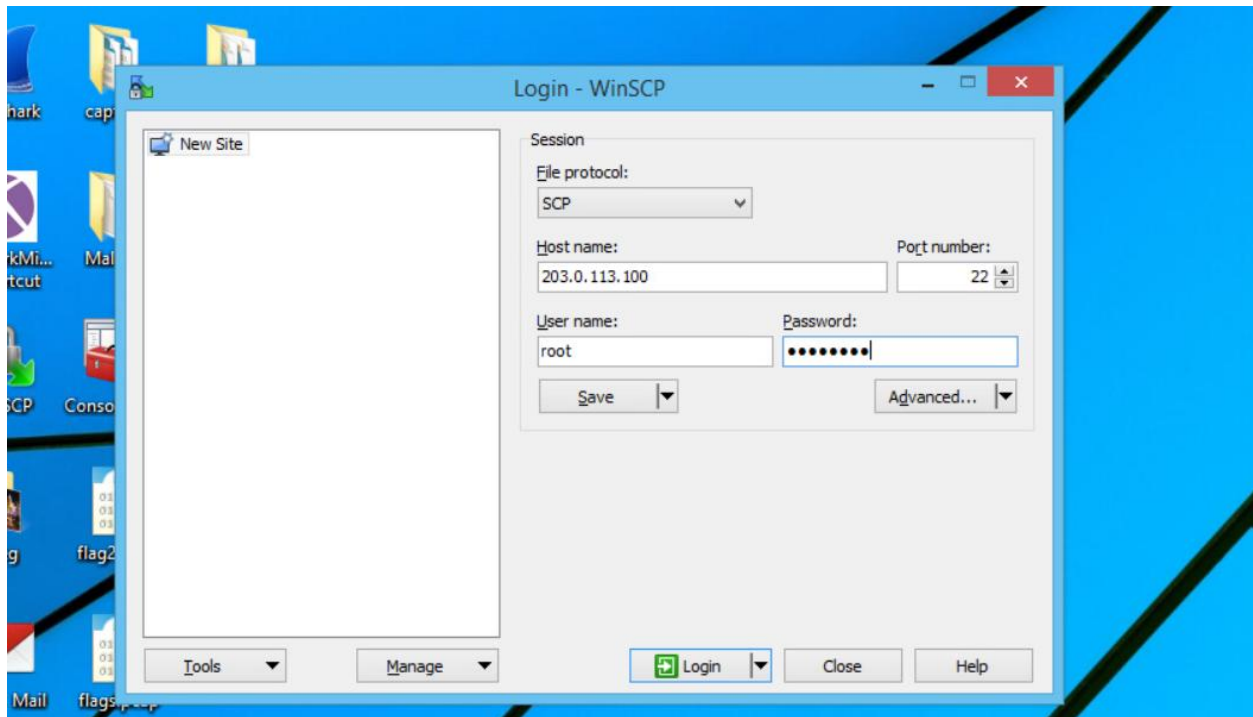
**PuTTY is a free and open-source terminal emulator, serial console, and network file transfer application. It is most commonly used as an SSH and Telnet client, allowing users to securely connect to remote servers, usually for administration or troubleshooting.**

8. Question 17

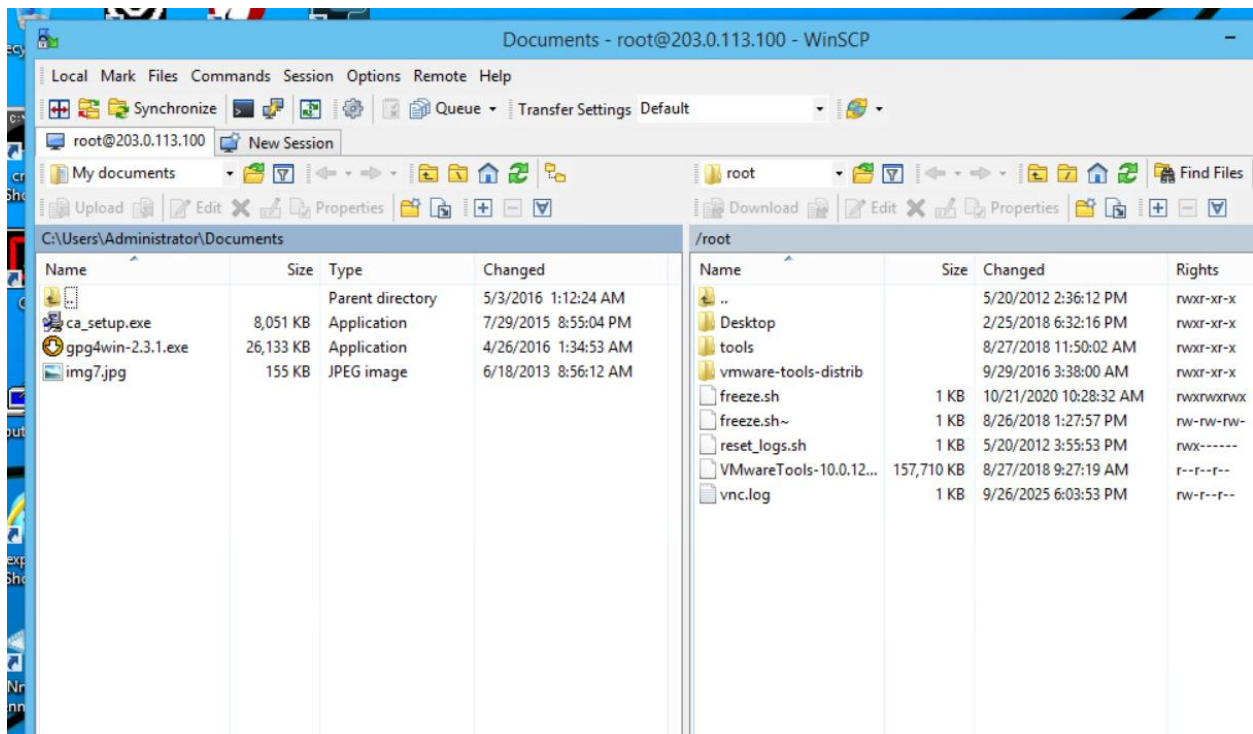




9. Question 26



10. Question 28



## Lab 2.1: Part 4 – Lessons Learned

Through this lab, I gained practical experience with key network security tools and firewall configurations. Using **Nmap and Zenmap**, I learned how to identify open ports and interpret scan results, which highlighted the importance of monitoring exposed services. I also explored **pfSense firewall rules**, specifically how to close unnecessary ports and create port forwarding rules, which reinforced how firewalls control network traffic and reduce attack surfaces. Additionally, I learned about **PuTTY** and its role in securely managing remote systems via SSH or Telnet. Overall, the lab demonstrated how scanning, firewall management, and secure remote access tools work together to strengthen network security.